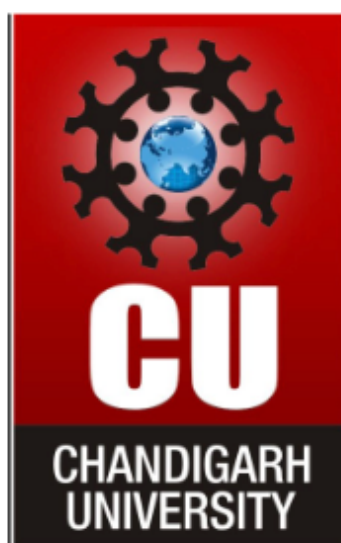


CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 7

**Subject: Advanced Database Management
System**

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1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:
 $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

Closure:

$B^+ = B$
 $BA^+ = BACD$
 $BC^+ = BCDA$
 $BD^+ = BDAC$

- Prime Attributes = {A,B,C,D}
- Super set / Candidate set = {AB, BC, BD}

Since, all the RHS attributes are prime, the relation R is in 3NF.

Check for BCNF:

For BCNF, for every non-trivial FD $X \rightarrow Y$, X must be a superkey.

$C \rightarrow D$, C is not a superkey, therefore it violates BCNF.

Highest Normal Form: 3NF

2. Relation R(ABCDE) having functional dependencies as :

$A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

$C^+ = C$
 $CA^+ = CADBE$
 $CB^+ = CBAED$
 $CD^+ = CD$
 $CE^+ = CE$

- Candidate sets = {CA, CB}
- Prime attributes = {A, B, C}

Since in $A \rightarrow D$, D can be determined by only A (subset of candidate key AC), it violates partial dependency so it is not in 2NF.

Highest Normal Form: 1NF

3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

$B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. Find the highest normal form.

Ans:

Closure:

$B^+ = BACDE$
 $A^+ = ACBED$
 $C^+ = C$
 $E^+ = E$

$$D^+=D$$

- Candidate keys: {A, B}
- Primary keys: {A, B}

Since all the attributes on left side are either candidate keys or super keys, hence the relation R is in BCNF form.

Highest Normal Form: BCNF

**4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:
 $A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$
 Find the highest normal form.**

Ans:

F never appears on the RHS of any FD, therefore it must be present in every candidate key.

$$\begin{aligned} AF^+ &= AFBCDE \\ BF^+ &= BFDACE \\ CF^+ &= CF \\ DF^+ &= DFABCE \\ EF^+ &= EF \end{aligned}$$

- Candidate keys: {AF, BF, DF}
- Prime attributes: {A, F, B, D}

A $\rightarrow$ C

A is a proper subset of candidate key AF, and C is non-prime.
 Therefore, there is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

**5. Relation R(ABCDE) having dependencies as:
 $CE \rightarrow D$,
 $D \rightarrow B$,
 $C \rightarrow A$**

Highest normal form? C.K set?

Ans:

Closure:

$$CE^+ = CEDBA$$

- Candidate key = {CE}
- Prime keys = {C, E}

From FD, $C \rightarrow A$

C is a proper subset of candidate key CE, and A is non-prime.
 Therefore, $C \rightarrow A$ is a partial dependency which violates 2NF.

Highest Normal Form: 1NF

6. Relation R(ABCDEF) having dependencies as:

AB → C

DC → AE

E → F

Highest normal form? C.K set?

Ans:

Since B, D never occur on RHS of any FD, they will be a part of candidate key.

Closure:

BD+=BD
BDA+=BDACEF
BDC+=BDCAEF
BDE+=BDEF
BDF+=BDF

- Candidate keys= {BDA, BDC}
- Prime attributes= {B, D, A, C}

Check for 2NF:

- DC → AE: A is prime but E is non-prime

Here, DC is a proper subset of candidate key BDC.

Therefore DC → E is a partial dependency so 2NF is violated.

Highest Normal Form: 1NF

7. Relation R(ABCDEFGHI) having dependencies as:

AB → C

BD → EF

AD → GH

A → I

Highest normal form? C.K set?

Ans:

Since A, B, D never occur on RHS of any FD, they will be a part of candidate keys.

Closure:

ABD+=ABDCEFGHI

- Candidate set = {ABD}
- Primary attributes = {A, B, D}

Check for 2NF:

- AB → C: AB is a proper subset of ABD and C is non-prime. Therefore:

AB → C is a partial dependency, so the relation violates 2NF.

Highest Normal Form: 1NF

8. Relation R(ABCDE) having dependencies as:

AB → CD

D → A

BC → DE

Highest normal form? C.K set?

Ans:

Since B doesn't appear on RHS, it must be the part of every candidate key.

Closure:

$BC^+ = BCDEA$
 $AB^+ = ABCDE$
 $BD^+ = BDACE$
 $BE^+ = BE$

- Candidate key = {BC, BA, BD}
- Prime attributes = {B, C, A, D}

Check for 3NF:

- **AB → CD**: AB is a super key
- **D → A**: D is not a super key but A is a prime attribute
- **BC → DE**: BC is a super key

Check for BCNF:

- **D → A**: D is not a super key

Therefore it violates BCNF

Highest Normal Form: 3NF

9. Relation R(ABCDE) having dependencies as:

BC → ADE

D → B

Highest normal form? C.K set?

Ans:

Since C does not appear on RHS, it must be a part of candidate key

Closure:

$C^+ = C$
 $BC^+ = ADEBC$
 $CD^+ = CDBAE$
 $CE^+ = CE$

- Candidate keys = {BC, CD}
- Prime attributes = {B, C, D}

Check for 3NF:

- **D → B**: D is not a super key, but B is a prime attribute
- **BC → ADE**: BC is a super key

Check for BCNF:

- **D → B**: D is not a super key, it violates BCNF

Highest Normal Form: 3NF

10. Relation R(ABC) having dependencies as:

A → B

B → C

C → A

Highest normal form? C.K set?

Ans:

Closure;

A+=ABC

B+=ABC

C+=ABC

- Candidate keys={A,B,C}
- Prime attributes={A,B,C}

Since all the LHS attributes are super keys,

Highest Normal Form= BCNF

11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 → F3

F2 → F4

(F1,F2) → F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Ans:

Since F1, F2 doesn't appear on RHS, they must be a part of candidate key.

Closure:

F1F2+=F1,F2,F3,F4,F5

- Candidate set={F1F2}
- Prime attributes= {F1, F2}

Check for 2NF:

- F1 → F3: F1 is a subset of candidate key F1F2 and F3 is a non-prime attribute, therefore it violates 2NF
- F2 → F4: F2 is a subset of candidate key F1F2 and F4 is a non-prime attribute, therefore it violates 2NF

Highest Normal Form: 1NF